18/07/23.

UNIT - 1

Classical Approaches of NLP

Computational Linguistics:- Language & Computer ⇒ ???

* explores how human language might be automatically processed & interpreted in a computational perspective.
* considers the mathematical & logical characteristics of natural language & develops algorithms & statistical processes for automatic language processing.
* Analysis focuses on anything from fundamental linguistic issues such as modelling the meaning of the word & recognising the grammatical structure of sentences to complex applications such as machine translation or assessment of statements for factual accuracy.
* analysis conducted using statistical & computational processes such as neural nws or processes borrowed from logic.
* CL ⇒ makes an contribution to further development of AI & serves as a driver of innovation in this field.
* multi-disciplinary experts work together ⇒ academic & industry linguists, computational linguistics, AIML investigators, cognitive scientists, speech specialists & philosophers.

Natural Language Processing (NLP):-

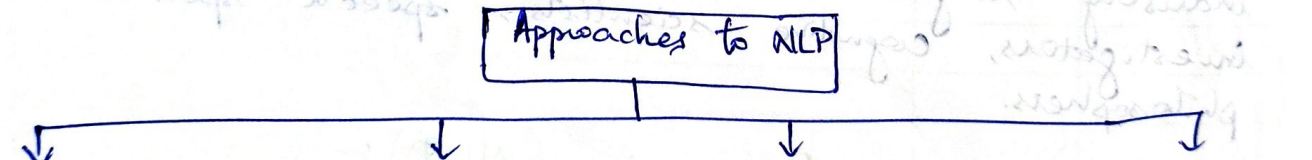
* application of computational techniques to the analysis & synthesis of natural language & speech.
* existing for more than 50 years for now, but still in research phase.
* branch of Computer Science & AI that combines computational linguistics - rule-based modeling of human language - with statistical, machine-learning & deep-learning algorithms.

* drives computer programs that
  → translate texts from one language to another
  → respond to spoken (voice) commands
  → summarize large volumes of text rapidly, even in real time.

Real-life applications of NLP:-

* voice-operated GPS systems (Google Maps)
* digital assistants (Siri, Alexa, Cortana, etc.,)
* speech to text dictation software (Google Docs voice typing, Apple Dictator, Speechnotes, etc.,)
* customer service chatbots (banking sector)
* text to text conversions (Google Translate)
* Auto-completion wizard (Whatsapp, Bing, Gmail, et
* Spelling & grammar checking tools (Grammarly, Ms-Office-Assistants)

Approaches to NLP:-



Perspectives Approaches to NLP

① Distributional approaches:-

* includes large-scale statistical tactics of ML & DL.
* turns content into word vectors for mathematical analysis & perform quite well at tasks such as.
  ⇒ POS tagging (is this a noun/verb?)
  ⇒ dependency parsing (does this part of sentence modify another part?)

⇒ semantic relatedness (are these words related (or) different words used in similar ways?)

* These NLP tasks don't rely on understanding the meaning of words, but rather on the relationship between words themselves.

Advantages:-

→ broad, flexible & scalable.
→ can be applied to different types of text without the need for hand-engineered features (or) expert-encoded domain knowledge.

Disadvantages:-

→ lack true understanding of real-world semantics & pragmatics.
→ achieve breadth, but not depth. Complex & nuanced questions that rely linguistic sophistication & contextual world knowledge have yet to be answered satisfactorily.

② Frame-based approaches:-

* frame → data structure for representing a stereotyped situation.
Eg:- "A commercial transaction" can be considered as a frame.

* Sentences that are syntactically different but semantically identical, can fit into same frame.
for instance: Cynthia sold Bob the bike for $200.
Bob bought the bike for $200 from Cynthia.

Disadvantages:- → require supervision ⇒ expert advisory (or) creation is needed, which limits the scope.
→ in many cases, incomplete.

(3) Model-Theoretical Approach:-

Model theory:-
* refers to idea to those instances referring real world. Eg:- the water is blue in oceans.

Compositionality:-
* meanings of parts of sentence can be combined to deduce the whole meaning.

What is the largest city in Europe by population?

⇓

max (Cities ∩ Containedby (Europe), population)

⇓

Istanbul

Advantages:-
* full-world representation
* rich semantics
* end-to-end processing

Disadvantages:-
* applications are heavily limited in scope due to the need for hand-engineered features.

(4) Interactive Learning:-

* an approach to tackle both breadth & depth in language learning is to employ dynamic, interactive environments where humans teach computers gradually.
* here, pragmatic needs of language inform the development.

Note:-
Syntax → what is grammatical
Semantics → what is meaningful
pragmatics → what is purpose/goal

Goals of Computational Linguistics:-

⇒ Theoretical goals:-

① formulation of grammatical & semantic for characterizing languages in ways enabling computationally tractable implementations of syntactic & semantic analysis.

② discovery of processing techniques & learning principles that exploit both structural & statistical properties of Language.

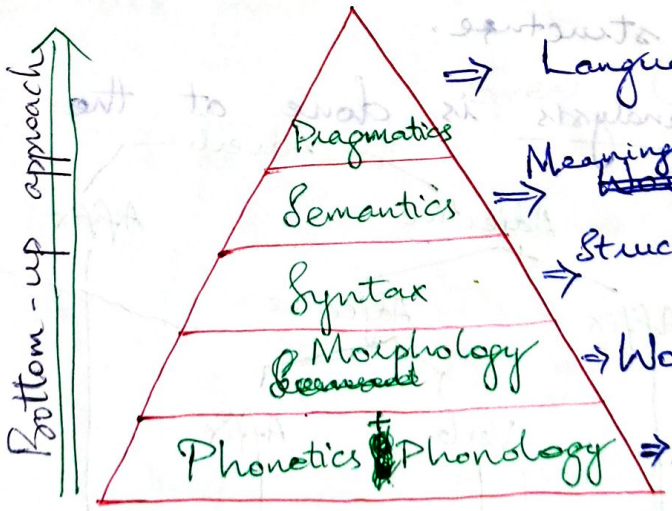
③ development of cognitively & neuroscientifically plausible computational models of how language processing might occur in brain.

⇒ Practical goals:-

* Creation of computational systems with human-like competency in dialogue, in acquiring language, & in gaining knowledge from text.

19/07/23.

Understanding linguistics → Levels in Linguistics analysis:-



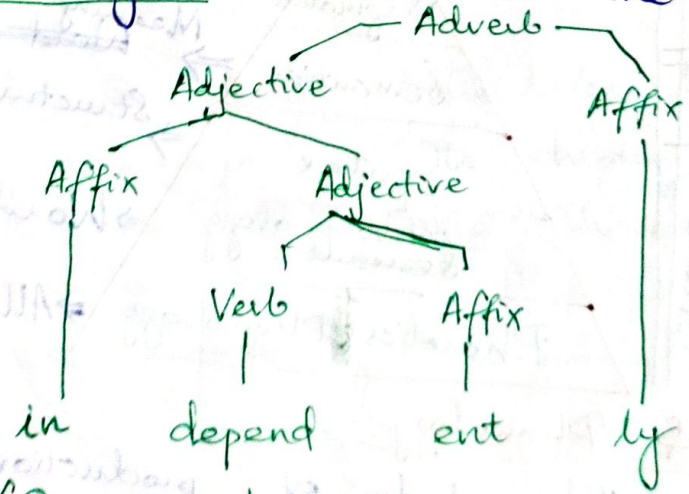
1) Phonetics & Phonology:-

* Phonetics is the study of production, transmission & perception of speech sound.
* Phonetics is related to the science of acoustics & is concerned with the sounds of languages, how these sounds are articulated & how the hearer perceives them.

* Phonology is the study of the patterns of a language, & is concerned with how sounds of languages are organized.
* It examines what occurs to speech sounds when they are combined to form a word & how these speech sounds interact with each other.
* NLP system accepts spoken input, sound wave & analyze it to encode into a digitized signal for interpretation.

2) Morphology :- (Basic building blocks)

* It refers to the word formation & mainly useful for identifying the parts of speech in a sentence & words that interact together.
* It also describes a set of relationship between words' surface forms & lexical forms.
* The info. gathered here at this stage prepares the data for the syntactical stage which looks more directly at the target language's grammatical structure.
* Morphological analysis is done at the word level.



3) Syntax :- (Grammatical representation)

* refers to the study of structural relationships between words in a sentence & involves applying the rules of Grammar.

* involves analysis of the words in a sentence to depict the grammatical structure of a sentence.
* it ~~aids~~ details an underlying structure of a transformational process.

Eg:- Alex drives the car.

Transformation ⇒ The car is ~~being~~ driven by Alex.

* Syntactic analysis is done at the statement level.

4) Semantics:- (meaning of the elements)

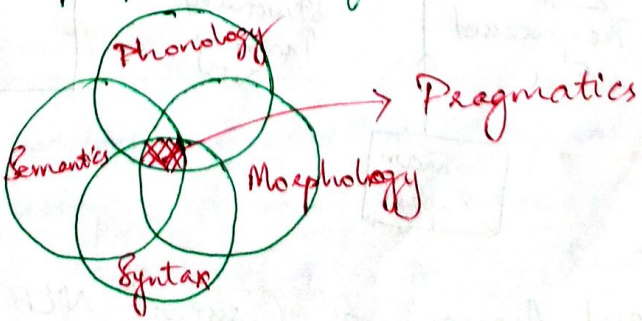
* deals with the examination of the meaning of words, phrases & sentences.
* determines the possible meanings of a sentence by focussing on the interactions among word-level meanings in the sentence.
* Semantic processing includes disambiguation of words with multiple senses.
* includes tasks like Named Entity Recognition (NER) & relationship extraction.

Semantically different:
(A) ~~How~~ How are you?
Great (B)

(A) How old are you?
20 years old (B)

Semantically similar:
(A) What's your age?
20 years old (B)

5) Pragmatics:- (purpose/goal of writing a sentence)



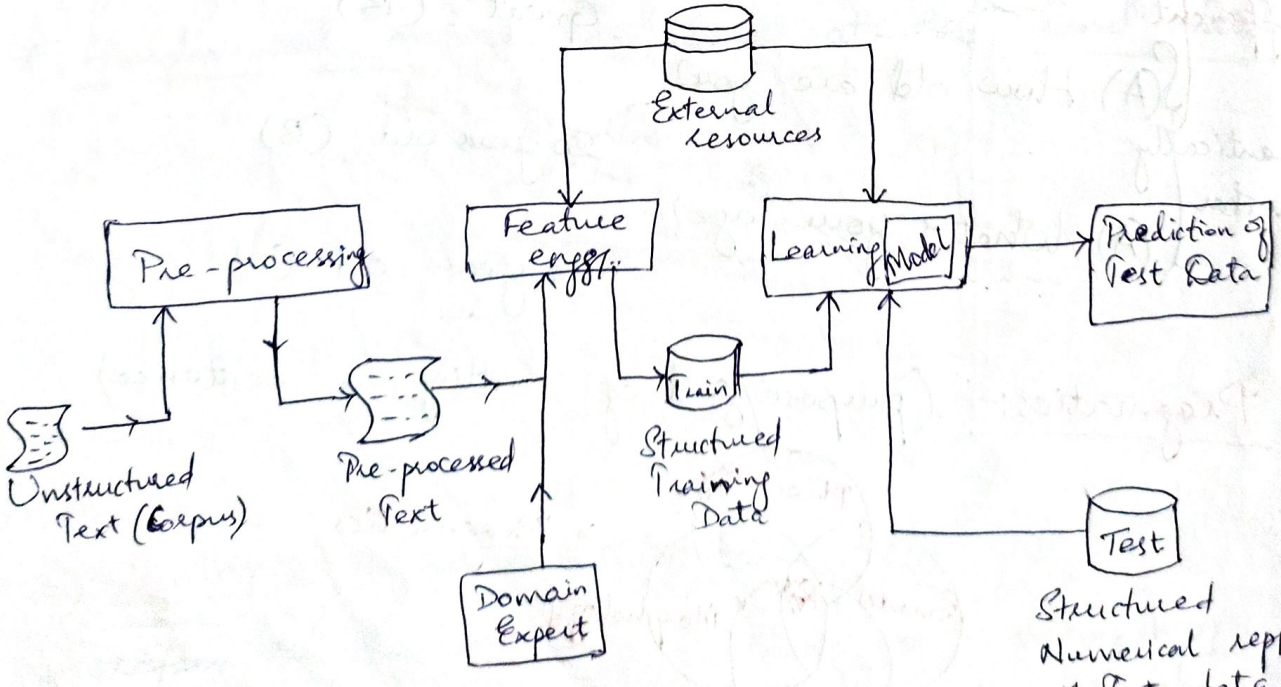
* analysis of the real meaning of an utterance in a human language, by disambiguating & contextualizing the utterance.
* Understanding the sentences created & to understand the conveyed meaning becomes the major goal of pragmatics.
* Common problems associated with pragmatics in the ML environment are:
  - → Co-referencing
  - → Summarization
  - → Modelling
  - → Question & Answering

28/09/23.

Traditional approach (or) General Sequential Flow in NLP:-

→ sequential flow of several key steps
1. Data Gathering → to gather the required I/P for a purpose
2. Pre-processing → removal of unwanted data.
3. Feature engg. → understand the numeral representation of textual data.
4. ML algorithms → learning the language using the training data.
5. Predicting outputs → identify the prediction with test data.

→ Statistical approach.



General Approach of Classical NLP

① Pre-processing:-

→ Text Corpora need to be pre-processed that focusses on reducing the vocabulary & distractions.

→ Distractions → things that distract the algorithm (punctuation marks, stop word removal, etc.,) from capturing the vital linguistic info. required for the task.

② Feature-Engineering:-

→ Objective ⇒ to make the learning easier for the algorithms.

→ most time-consuming & crucial step for obtaining good performance on a given NLP task.

→ Features are hand-engineered & biased towards the human understanding of a language.

→ Best ~~engine~~ performing systems has best engineered features.

→ Eg:- for a sentiment classification task, we can represent a ~~parse tree~~ sentence with a parse tree & assign positive, negative or neutral labels to each node / subtree in the tree to classify that sentence as positive or negative.

→ Additionally, this phase can use external resources such as Wordnet (a lexical database) to develop better features.

→ {Bag-of-words, n-gram} are some simple feature engg. techniques.

③ Learning Algorithm:-

→ learning algorithm learns to perform well at given task using obtained features & optionally, the external resources (eg:- in a text summarization task, a thesaurus that contains synonyms of words can be a good external resource.)

(4) Prediction:-

→ final stage & is very much straight-forward.

→ A new i/p is fed & obtain the predicted label by forwarding the i/p through the learning model.

Example - generating football game summaries:-

Task:- Consider a task of automatic text generation from the statistics of a game of football. We have several sets of game statistics (score, penalties, yellow cards, etc.,) & corresponding articles generated for that game by a journalist, as the training data.

Assumption:- for a given name, we have a mapping from each statistical parameter to the most relevant phase of the summary for that parameter.

Goal:- Given a new game, we need to generate a natural looking summary about the game. This can be as simple as finding the best-matching statistics for new game from the training data & retrieving the corresponding summary.

Step 1:- Pre-processing:

→ involves further steps as follows:

(i) Stemming/Lemmatization → choosing root verbs (converting listening to listen)

(ii) Removing distractions → eliminating all forms of punctuation

(iii) Tokenization → identifying simple words.

→ (i) & (ii) results in reducing the memory requirement.

→ Stemming & Lemmatization is a simple & trivial operation that relies on a simple set of rules, however it requires more than a simple rule base to develop a good stemming algorithm. Eg:- listened → listen ; agued → ague

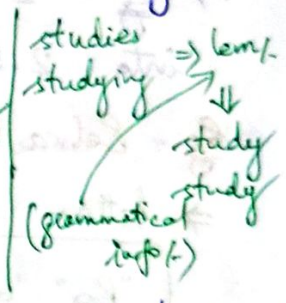
→ Effort can vary in complexity for other languages.

→ Tokenization → process of dividing a corpus into small entities (words).

→ Tokenization might appear ~~to~~ trivial for a language such as English, as the words are isolated; however not for certain languages like Thai, Japanese & Chinese, as these languages are not consistently delimited.

~~Step 2:- Tokenization~~

studies, studying ⇒ Stemming (removes suffixes) ⇒ studi, study



Step 2:- Feature Engineering

→ involves further steps as follows:

(i) Transformation → converting text data into numerical format

(ii) Representation → done using bag-of-words (or) n-gram (or) TF-IDF vector representations.

→ State-of-the-art classical models rely on much more sophisticated feature engg. techniques.

24/9/23.

Bag-of-words:-

→ creates feature representations based on word occurence frequency.

Eg:- ① Selva went to the shop to buy groceries

② Selva brought the groceries to give to his mother.

→ Vocabulary for these 2 sentences would be:

["Selva", "went", "to", "the", "shop", "buy", "groceries", "brought", "give", "his", "mother"].

→ to create a feature of size 'V' (vocabulary size) for each sentence showing how many times each word in the vocabulary appears in the sentence.

→ ① [1, 1, 2, 1, 1, 1, 1, 0, 0, 0, 0]

② ~~[1, 1, 1, 1, 2, 1]~~ [1, 0, 2, 1, 0, 0, 1, 1, 1, 1, 1]

⇒ limitation → it loses contextual info. as the order of words is no longer preserved, & hence useful inferences for downstream NLP tasks are not possible

n-gram:-

→ breaks down text into smaller components consisting of 'n' letters (or) words.

→ Eg:- 2-gram (bi-gram) would break the text into 2-letters (or) 2-words entities.

Eg:- Selva went to the shop to buy some flowers.

⇒ letter level 2-gram decomposition:

["Se", "el", "l", "v", "va", ..., "f", "fl", "lo", "ow", "we", "er", "rs"].

⇒ word level 2-gram decomposition:-

["Bob went", "went to", "to the", ... "buy some", "some flowers"]

→ Advantage of letter level ⇒ vocabulary will be significantly smaller than if we were to use words as features for large corpora.

3) Learning Process:- Data tuples format is used.

→ 3 sub modules: ① Hidden Markov model (HMM)
② sentence planner
③ discourse planner

→ HMM learns the morphological structure & grammatical properties in the language by analyzing the related phrases in the text corpus.

→ Sentence planner corrects any linguistic mistake whether it comes as grammatical one (or) a morphological one in the phrases.

→ Discourse planner orders & structures the set of sentences in the correct format for a proper meaning to be conveyed.

④ Prediction:-

→ Test Data is provided to the algorithm for prediction & the results are gathered.

Drawbacks:-

① loss of info.. while reducing the vocabulary..
② Feature engg.. is a lengthy & complicated process.
③ Domain expertise & external resources may not be readily available.
④ Creation & identification of external resource might be painstaking..

Statistical Methods:-

① Word frequency:-

→ most commonly used.

→ used to identify the most regularly used expressions (or) words that is present in a specific text.

→ this freq.. provides insights on usage pattern of any natural language & can be useful in categorizing elements.

② Collocation:-

→ to identify the common words that appear together, & these are co-occuring words like bigrams (take off) & trigrams (see you later).

③ Concordance:-

→ methodology to provide context to the natural language.

→ Contextual meaning of a single word might lead to different ways in which the word can be understood.

Eg:- book ⇒ * I read a book about the history of the Cholas.

* I booked a ticket in the app.

④ TF-IDF:-

→ Term Frequency - Inverse Document Frequency identifies/measures how important a term is within a document relative to collection of documents (corpus).

→ Usage of the word can be unique to a document (or) usage can be pretty common across multiple docs.. (eg.. in, is, are, on, the, etc.)

$$TF = \frac{\text{no}/\ldots \text{ of times term 't' appears in doc/text}}{\text{total no}/\ldots \text{ of terms in doc/text}}$$

$$IDF = \log_2 \left(\frac{\text{Total no}/\ldots \text{ of doc}/\ldots}{\text{no}/\ldots \text{ of doc}/\ldots \text{ with term 't' in it}} \right)$$

POS Tagging:-

→ process of making-up a word in a text (corpus) as corresponding to a particular part of speech, based on both its definition & its context.

→ In NLP, POS Tagging is a fundamental task which involves assigning a grammatical category such as Noun, Verb, etc. to each word in a sentence.

→ goal is to understand the syntactic structure of a sentence & identify the grammatical roles of individual words.

→ Penn Treebank POS tag set is widely used in English NLP tasks.

Importance of POS Tagging:-

→ syntactic analysis (to tag the POS)

→ semantic analysis (to understand meaning of words in a content without ambiguity)

→ information retrieval (to improve precision & relevance of search results)

→ Named Entity Recognition (NER) tasks

→ Machine Translation

Example of POS Tagging:-

The	quick	brown	fox	jumps	over	the	lazy	dog.
DT	JJ	JJ	NN	VBZ	IN	DT	JJ	NN

POS Tags from Penn Treebank :-

CD → cardinal no. (one, two, ---)
DT → determiner (a, an, the)
FW → foreign word (rendezvous)
IN → preposition (in, over, at, ---)
JJ → adjective (red, beautiful, ---)
JJR → comparitive adjective (bigger, smaller, ---)
JJS → superlative adjective (biggest, smallest, ---)
MD → modal (can, should, ---)
NN → singular / mass noun (fox, dog, bottle, ---)
NNP → proper noun, singular (Jack, IBM, ---)
NNPS → proper noun, plural (Carolinas, ---)
NNS → plural / mass noun (foxes, dogs, ---)
PRP → personal pronoun (I, He, She, ---)
RB → adverb (fastly, slowly)
TO → to (to)
VB → verb base (eat, run, ---)
VBD → verb, past tense (ate, ran, ---)
VBG → verb, gerund (eating, running, ---)
VBZ → verb, 3rd person, singular, present tense (eats, runs, ---)

Types of POS Tagging :-

① Rule - based tagging,
② Transformation - based tagging, &
③ Statistical - based tagging.

① Rule-Based POS Tagging:-

→ involves assigning words with their respective POS using pre-determined rules, that are based on specific word characteristics & contextual cues.

→ Eg:- Assign the POS tag 'NN' to words ending in '-tion' or '-ment'.

The	presentation	highlighted	the	key	achievements	of	the
DT	NN	VBD	DT	JJ	NNS	IN	DT

project's	development.
NN	NN

Knowledge-driven Taggers

② Statistical-based POS Tagging:-

→ also called as stochastic process / probabilistic process of POS tagging, where algorithms learn the probabilistic distribution of word-tag sequences.

→ Conditional Random Field (CRF) & Hidden Markov Models (HMM) are popular models, where it estimates the chances of observing a specific tag given the current word & its context by learning from labelled samples during training.

③ Transformational-based POS Tagging:-

→ a combination of probabilistic & rule-based approaches to automatically calculate symbolic rules from a corpus.

10/08/23.

Hidden Markov Model (HMM):-

→ well-known model based on probabilistic calculations.

→ effective in language processing activities, such as POS tagging of a sentence, etc., thereby predicting the most probable state sequence for any sentence.

→ takes greedy decisions as it follows brute-force approach.

Need for HMM:-

Eg:- Aim:- to tag the below sentence with POS.

① May saw Will.

Steps involved:-

(i) Collect data → labelled data

(ii) Create lookup table → tag each word with most common POS.

(iii) Tag the target sentence → while going word by word, tag the most common POS associated with it.

(i) Data Collection & labelling:-

Jane	saw	Will
May	saw	Jane
Noun	Verb	Noun

(ii) Look-up table:-

	N	V
May	①	0
saw	0	②
Will	①	0
Jane	②	0

(iii) Tag the target sentence:-

Target → May saw Will

Based on most common POS association,

May → Noun

saw → Verb

Will → Noun

(2) Mary will see Will. → tag it

(i) Labelled data:-

Based on datasets, classifying / labelling them,

Mary	will	see	Jane.
Will	will	see	Mary.
Jane	will	see	Will.
Noun	Modal	Verb	Noun

(ii) Lookup table:-

	Noun	Verb	Modal
Mary	(2)	0	0
see	0	(3)	0
Jane	(2)	0	0
Will	2	0	(3)

⇒

(iii) Tag the target sentence:-

Mary → noun

will → modal

see → verb

Will → modal X noun?

From the above example, its evident that, this mtdy. of tagging won't work for all the cases. Look-up tables won't work, if a single word can have multiple tags, ie. it won't consider any context.

Here, we try using bigrams & altering the table:

	N-M	M-V	V-N
may - will	(1)	0	0
will - see	0	(3)	0
see - jane	0	0	(1)
will - will	(1)	0	0
see - mary	0	0	(1)
jane - will	(1)	0	0
see - will	0	0	(1)

⇒

Mary will see Will.

N M/M V/V N.

③ Jane will spot Will. → tag it.

Labelled data is as follows:

→ Mary(N) Jane(N) can(M) see(V) Will.(N)

→ Spot(N) will(M) see(V) Mary.(N)

→ Will(M) Jane(N) spot(V) Mary(N) ?

→ Mary(N) will(M) pat(V) Spot.(N)

In the given dataset, combination / bigram of Jane-will is not available & hence to solve these kind of cases, HMM - Hidden Markov model is used.

In HMM, we will use 2 probabilities:

① Transition probability :-

it determines how likely is that, the noun followed by a modal which is followed by a verb & then a noun.

(changes in the states ~~in~~ an i/p ⇒ noun → modal → verb → noun ...)

② Emission probability :-

it determines how likely are the words with ambiguities are noun (or) modal (or) verb, etc.

(will be a modal / noun ; spot ~~will~~ a verb / noun)

Look-up table for Emission probability :-

	noun	modal	verb
mary	4/9	0	0
jane	2/9	0	0
will	1/9	3/4	0
spot	2/9	0	1/4
can	0	1/4	0
see	0	0	2/4
pat	0	0	1/4
	9	4	4

Look-up table for Transition probability :-

To obtain more clarity on transitions, we are going to add 2 more tiles/tags → start & end tags, for the dataset.

For eg., Doc. 1 ⇒ <S> Mary Jane can see Will <E>
(N N M V N)

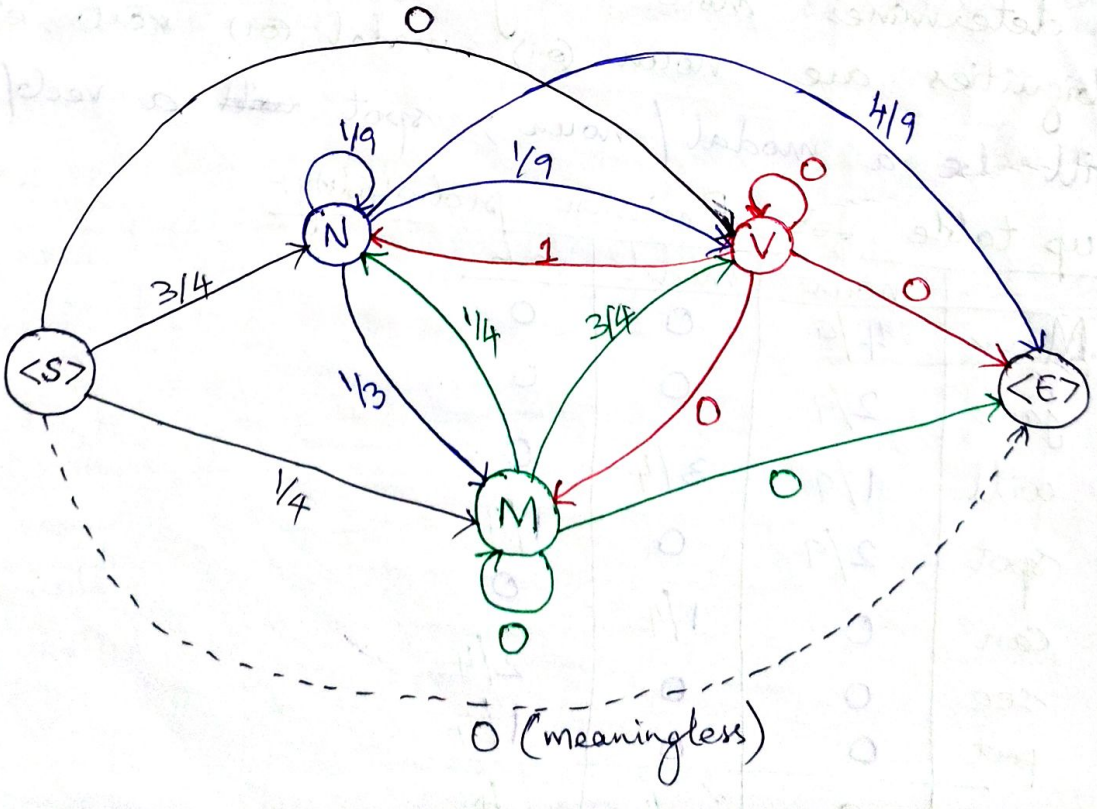
Doc. 2 ⇒ <S> Spot will see Mary <E>
(N M V N)

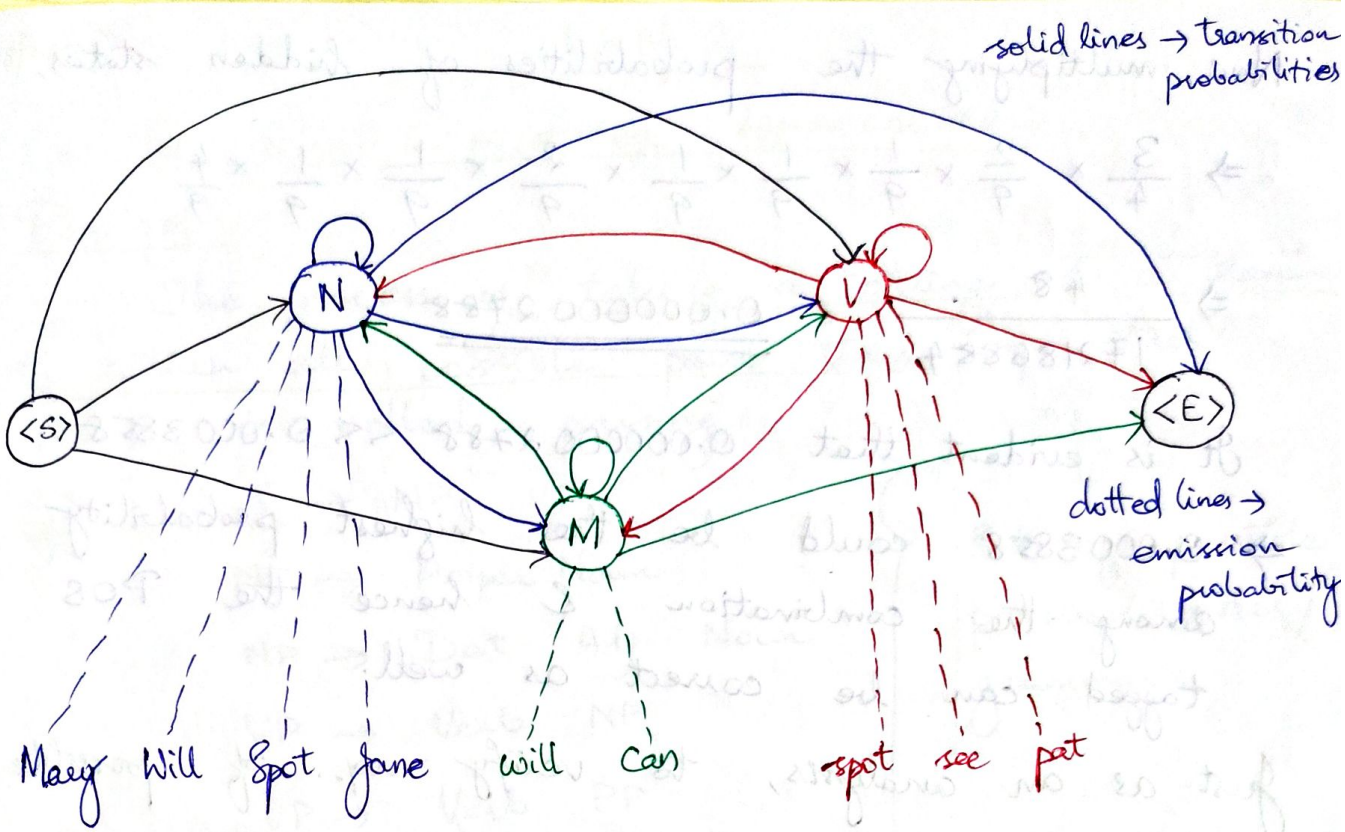
Doc. 3 ⇒ <S> Will Jane spot Mary <E>
(M N V N)

Doc. 4 ⇒ <S> Mary will pat Spot <E>
(N M V N)

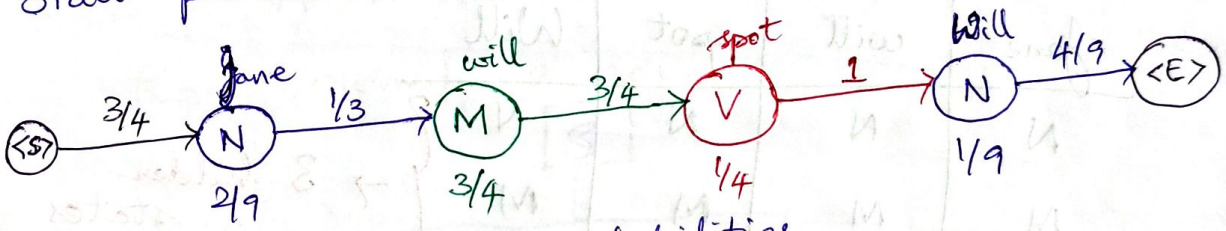
	N	M	V	<E>	
<S>	3/4	1/4	0	0 →	4 meaningless
N	1/9	3/9 = 1/3	1/9	4/9	9
M	1/4	0	3/4	0	4
V	4/4 = 1	0	0	0	4

State diagram for transition probabilities looks like below :





Start predictions,

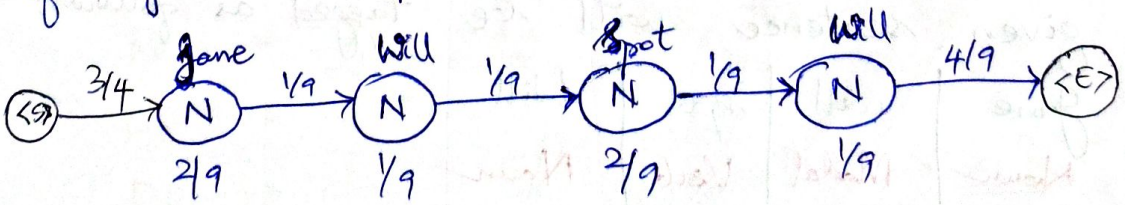


Now, multiplying all probabilities,

$$\Rightarrow \frac{3}{4} \times \frac{2}{9} \times \frac{1}{3} \times \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} \times \frac{1}{1} \times \frac{1}{9} \times \frac{4}{9}$$

$$\Rightarrow \frac{216}{559872} = \underline{\underline{0.0003858}}$$

We perform multiplication of probabilities as each & every probability is independent of all the other events. Since, how likely that start word is a noun has absolutely no effect (or) how likely noun will be followed by modal, verifying this by identifying the hidden states & finding the probabilities,



Now, multiplying the probabilities of hidden states,

$$\Rightarrow \frac{3}{4} \times \frac{2}{9} \times \frac{1}{9} \times \frac{1}{9} \times \frac{1}{9} \times \frac{2}{9} \times \frac{1}{9} \times \frac{1}{9} \times \frac{4}{9}$$

$$\Rightarrow \frac{48}{172186884} \quad \therefore = \underline{\underline{0.000000 2788}}$$

It is evident that $0.000000 2788 << 0.000 3858$

$\Rightarrow$ 0.000 3858 could be the highest probability among the combination & hence the POS tagged can be correct as well.

Just as an analysis, to verify no. of possible probabilities,

Jane	will	spot	Will
N	N	N	N
M	M	M	M
V	V	V	V

→ 3 hidden states

$\Rightarrow 3^4 = 81$ possibilities.

Disadvantages:-

→ hidden states & data size grows & hence very complicated to compare & predict the largest probability.

$\Rightarrow$ the above disadv. can be overcome by Viterbi Algorithm.

Conclusion:-

The given sentence will be tagged as follows:

Jane	will	spot	Will
Noun	Modal	Verb	Noun

Viterbi Algorithm:-

→ It is a dynamic programming algorithm which is widely used for finding the most likely sequence of hidden states in a HMM, called the Viterbi path, which results in a sequence of observed events, especially in the context of POS tagging.

→ It involves 2 basic steps:

i) Forward passing

ii) Back-tracing.

→ Back-tracing in the Viterbi algorithm is the process of tracing the path of POS tags that led to the highest probability for the entire sequence.

→ This process uses the backpointer table built during the forward phase of the algorithm & results in the most-likely sequence of tags for the given sentence.

→ This algorithm overcomes the major limitation of HMM-based POS tagging, ~~there~~ arising due to the exponential growth of hidden states which results in time & space complexities.

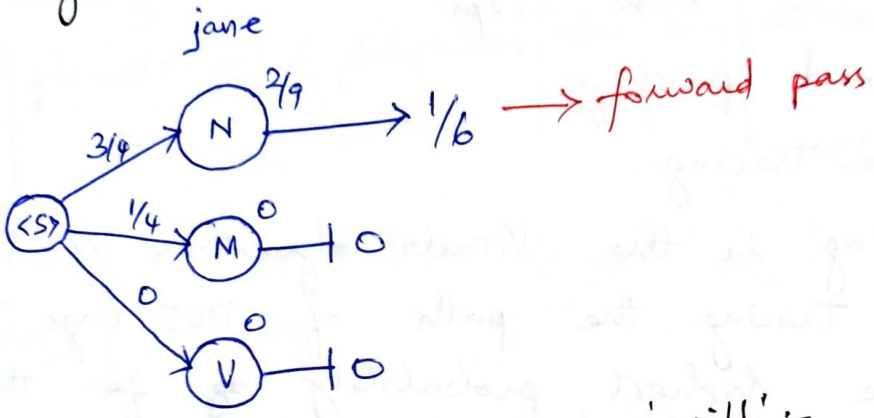
→ Instead of making greedy decisions at each step (brute force mtds.), the Viterbi algorithm results in obtaining global optimum with better performing in the case of time & space complexities.

Example :-

Considering the same example under HMM, after calculating the transition & emission probabilities, we can start the predictions:

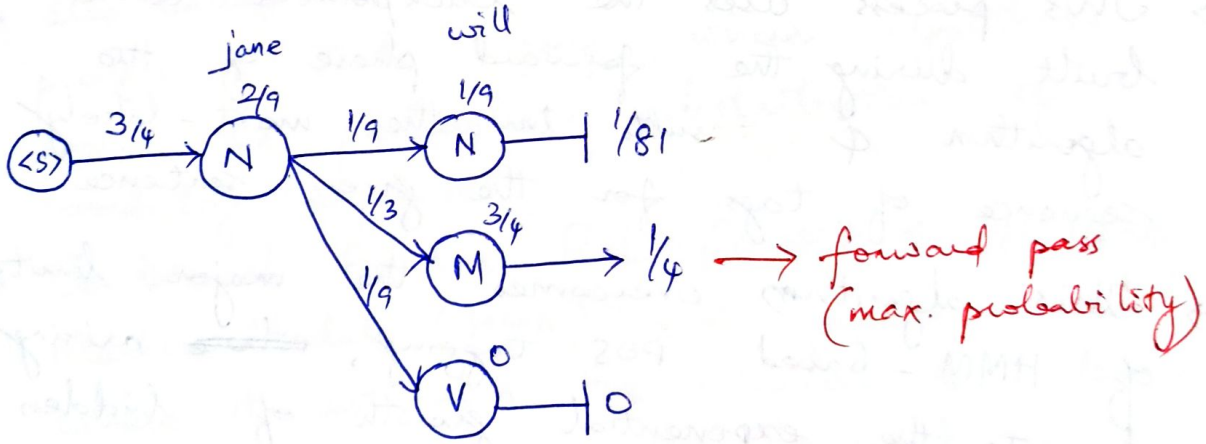
To tag: jane will spot will.

(i) Finding the POS tag for 'jane' :-



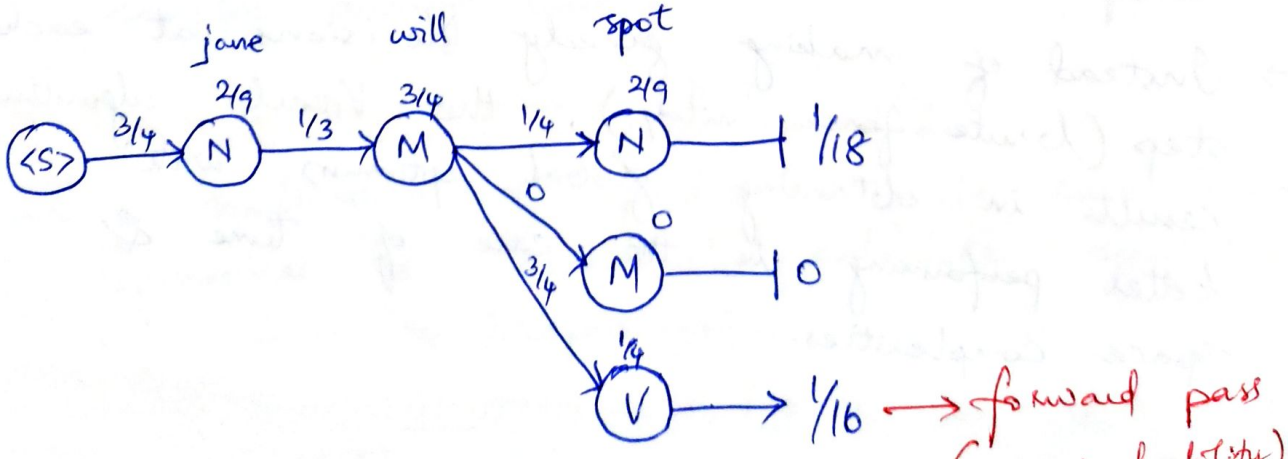
→ forward pass

(ii) Finding the POS tag for 'will' :-



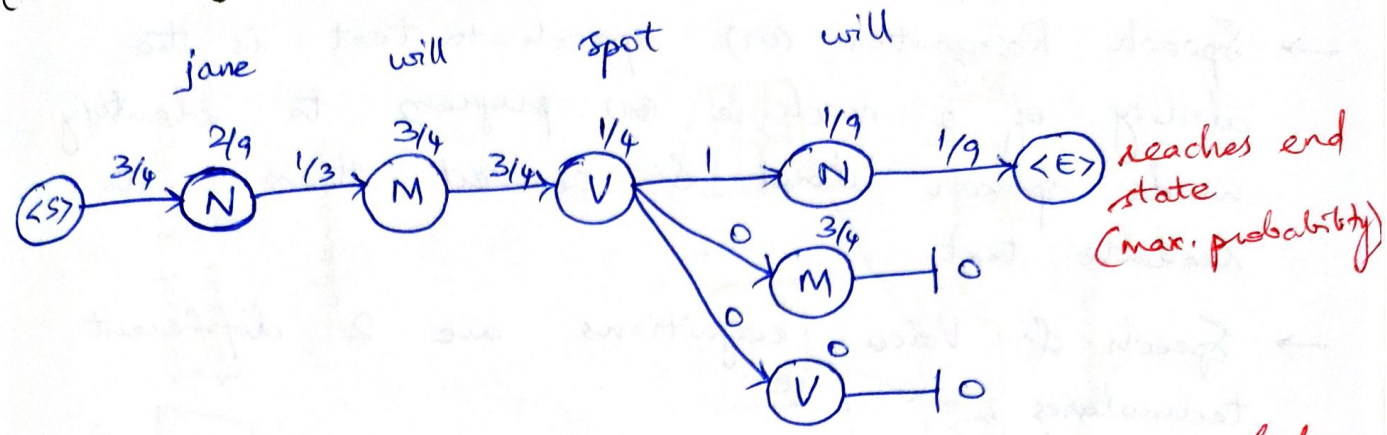
→ forward pass (max. probability)

(iii) Finding the POS tag for 'spot' :-



→ forward pass (max. probability)

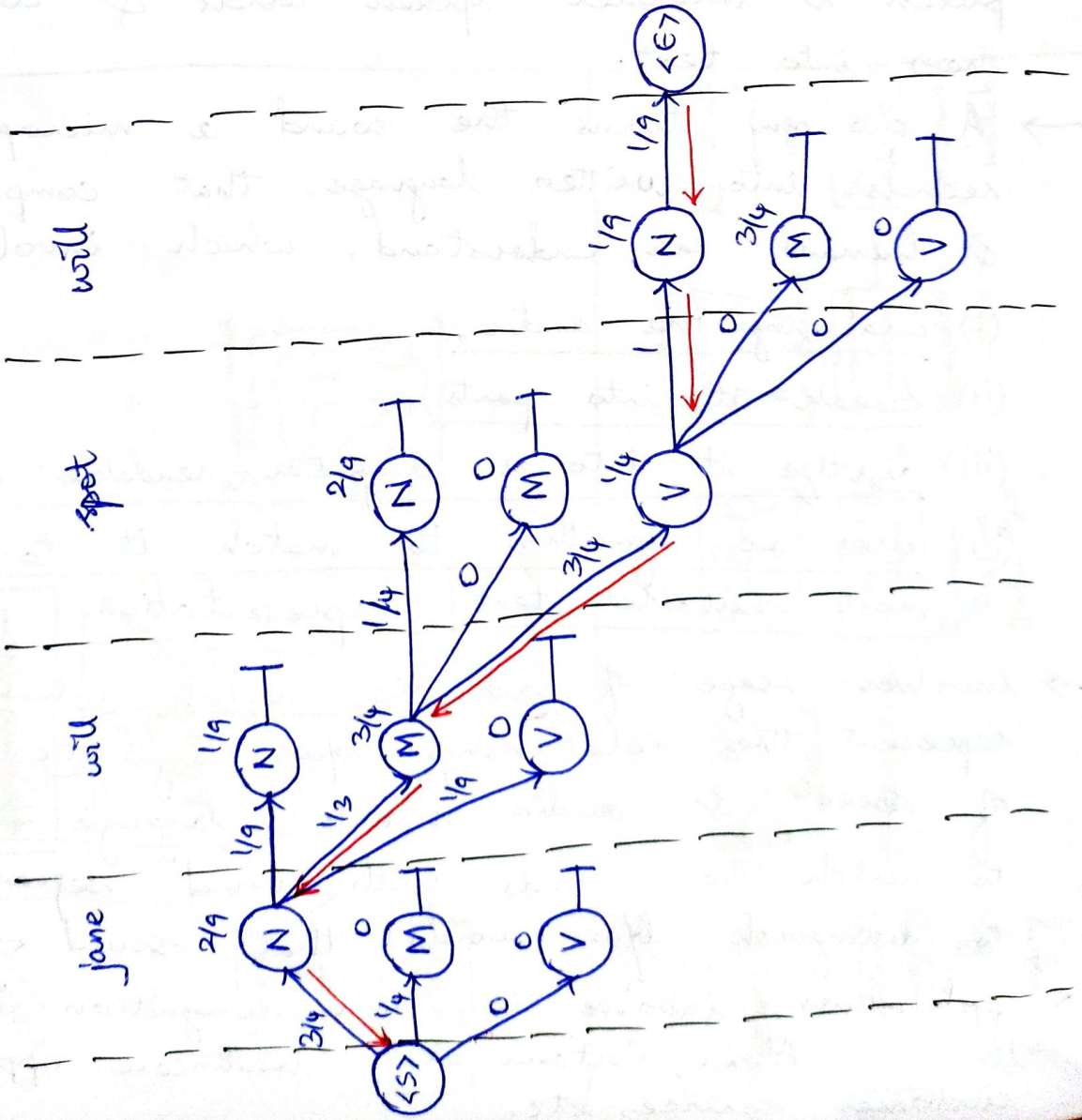
(iv) Finding the POS tag for 'will':-



The actual state diagram will look as per below. Upon back-tracking on the Viterbi path (the path which provided the maximum likelihood probability based on HMM chain), we can conclude & tag the i/p sentence as follows:

jane	will	spot	will.
Noun	Modal	Verb	Noun

State Diagram & Viterbi path



Probabilistic Speech Recognition:-

→ Speech Recognition (or) speech-to-text is the ability of a machine (or) program to identify words spoken aloud & convert them into readable text.

→ Speech & Voice recognitions are 2 different technologies:

* Speech recognition identifies words in a spoken language.
* Voice recognition is a biometric technology for identifying an individual's voice.

Basic principle of speech recognition:-

→ these systems use computer algorithms to process & interpret spoken words & convert them into text.

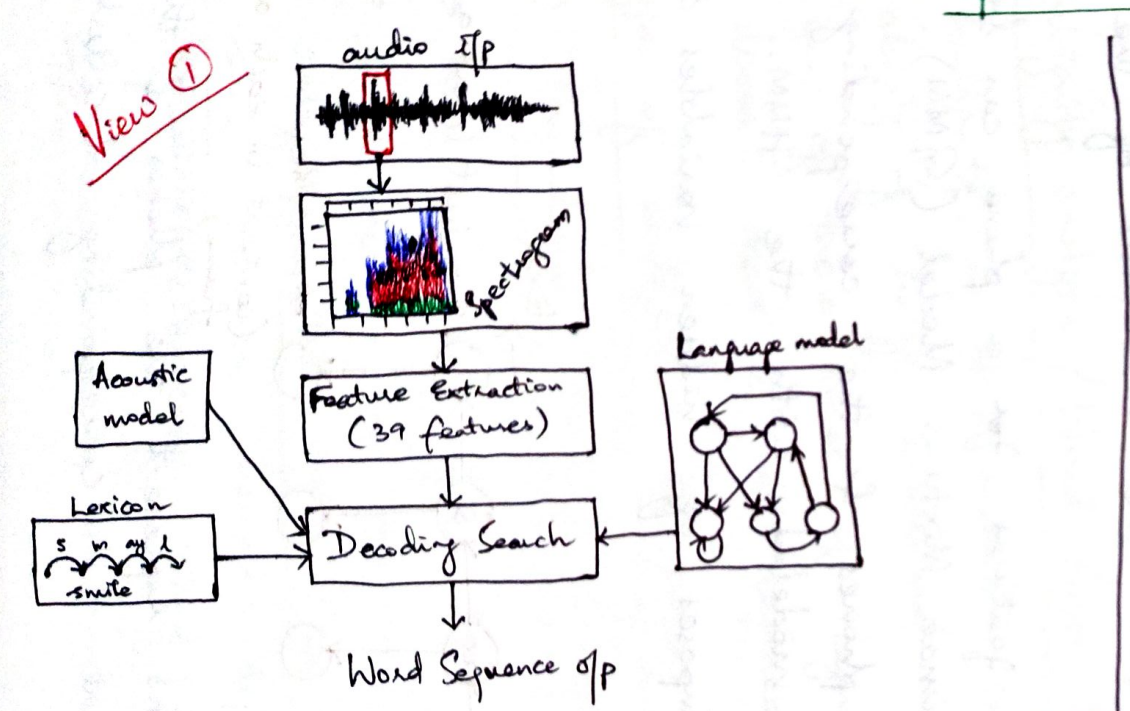
→ A s/w pgm.. turns the sound a microphone records into written language that computers & humans can understand, which involves:

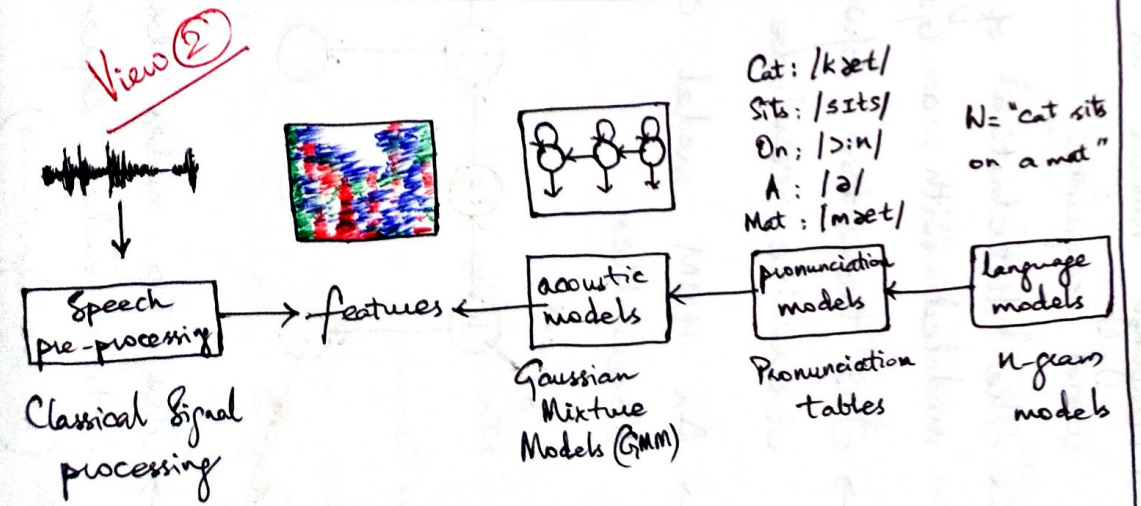
(i) analyzing the audio,
(ii) break it into parts,
(iii) digitize it into a computer-readable format,
(iv) uses an algorithm to match it to the most suitable text representation.

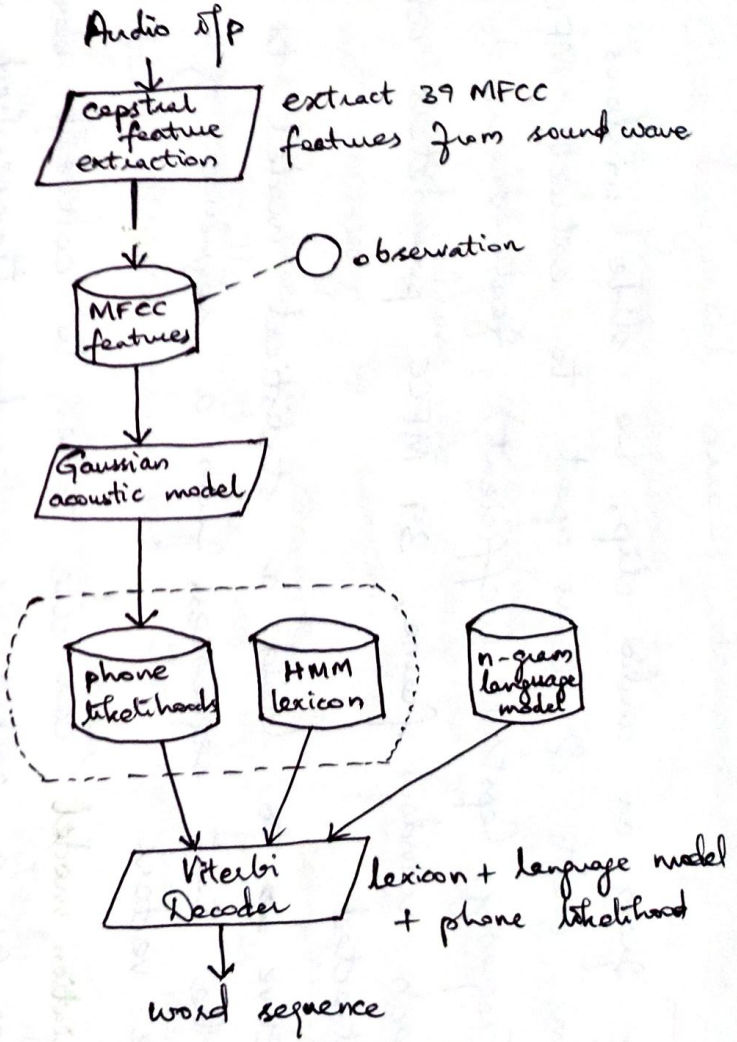
→ involves usage of acoustic models which represent the relationship b/w linguistic units of speech & audio signals; language models to match the sounds with word sequences to distinguish b/w words that sound familiar.

→ applications involve in voice-recognition applications like Siri, Alexa, Cortana, etc., healthcare applications, customer service, etc.

Architecture Diagram / Flow of HMM-Based Speech Recognition







Working:-

→ Starting from an audio clip, we slide windows of 25ms width & 10ms apart to extract MFCC (Mel Frequency Cepstral Coefficient) features.

→ For each window frame, 39 MFCC parameters will be extracted.

→ 1° objective → to build a statistical model to infer the text sequences, from a sequence of feature vectors

→ Pronunciation model can use tables to convert words to phones, (or) a corpus is already transcribed with phonemes already.

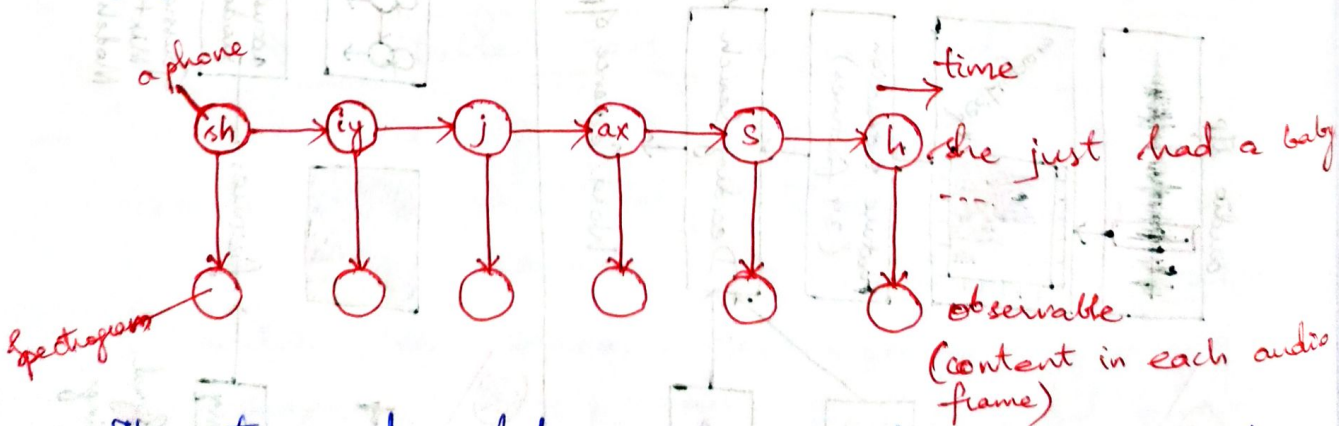
→ Acoustic model is about modeling a sequence of feature vectors given a sequence of phones instead of words.

→ Language model is about the likelihood of the word sequence.

→ The distribution of features for a phone can be modeled with a Gaussian Mixture Model (GMM).

→ The transition b/w phones & the corresponding observable can be modeled with the HMM.

→ An HMM model composes of hidden variables & observables.



→ The top nodes below represent the phones & the bottom nodes represent the corresponding observables (audio features).

→ The horizontal arrows demonstrate the transition in the phone sequence for the true label "she just ...".
→ observable is represented by 39 MFCC features extracted from the corresponding audio frame.
→ not necessary to search the phone sequence one-by-one, as otherwise, the complexity grows exponentially with the no.. of phones.
→ With Viterbi algorithm, the optimal sequence can be found in polynomial time.
→ The decoding is done through Viterbi decoder, that finds the most likely sequence of hidden states by computing the probabilities of all possible sequences of hidden states & selecting the one with the highest probability.

Ambiguity analysis / Word Sense Disambiguation :-

→ Ambiguity, in NLP contexts, can be referred as the ability of being understood in more than one way.
→ one of those areas of cognitive sciences that doesn't have a well-defined solution.

Types of ambiguities in NLP:-

① Phonological / Phonetical ambiguity:-

→ used only for speech processing & not texts.
→ 2 (or) more words are said to possess phonetical ambiguity, when their sounds of pronunciation are same, however having different spellings & meanings (technically called as homophones).
Eg:- weak - week / cell - sell / two - to - too, etc.

② Morphological ambiguity:-

→ arises when a word can be analyzed in more than one way due to its morphological structure, ie. the way in which the word is formed from its base components (roots, prefixes, suffixes).

→ Eg:- unlockable ⇒ un + lock + able

Interpretation ①:- something that can be unlocked (un- + lockable)

Interpretation ②:- something that cannot be unlocked (unlock + -able).

③ Lexical ambiguity:-

→ occurs when a single word has more than one meaning, regardless of its morphological structure.

→ associated purely with a lexicon (vocabulary of a language).

→ can arise in 2 different scenarios:

(i) a lexicon can have same category of parts of speech, however meanings vary.

Eg:- bank → POS is noun.

⇒ interpretations are financial institution & side of a river/stream.

(ii) a lexicon can have different meanings due to the usage variation in different POS.

Eg:- book → as noun/verb.

⇒ interpretation as noun → physical copy of printed pages bound together. Eg:- I am reading a book.

⇒ interpretation as verb → to reserve/arrange things in advance. Eg:- She booked tickets for a movie.

(4) Syntactic ambiguity:-

→ also known as structural ambiguity, occurs when a sentence can be parsed / structured in more than one way due to its syntax (sentence structure).

→ ambiguity arises because the sentence can be interpreted with different syntactic structures.

Eg:- The burglar threatened the man with the knife.

⇒ This sentence can be parsed in 2 different ways as shown below:

(i) The prepositional phrase, "with the knife" can be attached to the lexicon "man" to be interpreted as "the burglar threatened the man who had a knife".

(ii) The prepositional phase "with the knife" can be attached to the lexicon "burglar" to be interpreted as "the burglar threatened the man using a knife".

(5) Semantic ambiguity:-

→ occurs when a word, phase (or) a sentence has more than one meaning, because of multiple interpretations.

Eg:- ① I saw her duck.

interpretation ① ⇒ the speaker saw a duck that belonged to her (where "duck" is a noun).

interpretation ② ⇒ the speaker saw that she lowered her head (where "duck" is a verb → means to lower one's one head / body).

② The car hit the pole while it was moving.

interpretation ① ⇒ the car while moving could have hit the pole (fixed), causing an accident.

interpretation ② ⇒ the car got hit (no movement), while the pole was moving.

⑥ Pragmatic ambiguity :-

→ arises from the context in which a sentence/phrase is used that refers to the processing of user intention, modelling, in particular from the speaker's intent, the listener's interpretation, (or) the situation in which the communication occurs.

→ deals with the practical aspects of the meaning.

Eg:- Can you pass the book?

interpretation ① ⇒ a literal question asking if the person is able to pass the book.

interpretation ② ⇒ a polite request for the person to pass the book.

Common NLP techniques for Text Extraction/Classification:-

① Named Entity Recognition (NER):-

→ a task that is concerned with identifying & classifying named entities in textual data.
→ Named entities can be a person, organization, location, date, time (or) even quantity.

② Key topic / Key word extraction:-

→ focusses on extracting key topics / words discussed in a given i/p text.
→ usually much more refined key phrases (or) sentences instead of one word keywords & list of broad topics.

③ Sentiment Analysis:-

→ to understand the emotional feel in a textual content.
→ categorization for instance can be +ve, -ve (or) neutral.
→ Fields → social media monitoring, market research, customer support, product reviews, etc.

④ Clustering:-

→ to sort through large quantities of unstructured data.
→ Eg:- Search engines use clustering for selecting up the terminologies searched for in few seconds.

Text Classification & Extraction:-

→ Text classification is the process of labelling or organizing text data into groups.

→ Content is analyzed & classified into multiple pre-defined groups based upon the analysis.

→ Majority of the data is unstructured & hence classifying this data can be extremely useful.

→ Text extraction is the process of gathering valuable pieces of information present within the text & data.

Applications of Text classification:-

* Spam detection in emails
* Sentiment analysis of online reviews
* Topic labelling docs. like research papers
* Language detection (like in Google Translate)
* Age/gender identification of anonymous users
* Tagging online content
* Speech recognition used in virtual assistants like Siri & Alexa.

Approaches of Text Classification:-

1) Rule-based approaches
2) Machine learning approaches
3) Hybrid approaches

① Rule-based approaches:-

→ make use of handcrafted linguistic rules to classify text.

→ One way to ~~create~~ group text is to create a list of words related to a certain column & then judge the text based on the occurences of these words.

Eg:- fur, feathers, claws, scales → about animals

Donald Trump, Boris Johnson → politics

palace, castle, park → sight-seeing / architecture

→ approaches require a lot of domain knowledge to be extensive, take a lot of time to compile & are difficult to scale.

2) Machine learning approaches:-

→ using ML to train models on large sets of text data to predict categories of ~~text~~ new text.

→ To train models, we need to transform text data into numerical data → feature extraction. (bag-of-words & n-gram).

→ Several useful ML algorithms for text classification are,

* Naive Bayes classifiers
* Support Vector Machines
* Deep Learning algorithms

3) Hybrid approaches:-

→ combination of ① & ②, to model a classifier that can be fine-tuned in certain scenarios.

Information Retrieval

Importance of moving beyond Keyword Search in IR processes:-

→ While keyword search serves as a fundamental approach to IR, it does not consider the importance of terms within a document (or) across the document collection.

→ To address these limitations, & improve search accuracy, more advanced techniques like TF-IDF & Vector Space models are used.

Concept of TF-IDF (Term Freq. - Inverse Document Freq.):

→ main idea behind TF-IDF is that the terms that appear frequently in a document, but rarely across the entire document collection are more significant & informative.

Term Frequency (TF) is the ratio of no. of times term 't' appears in a document to that of total no. of terms in the document.

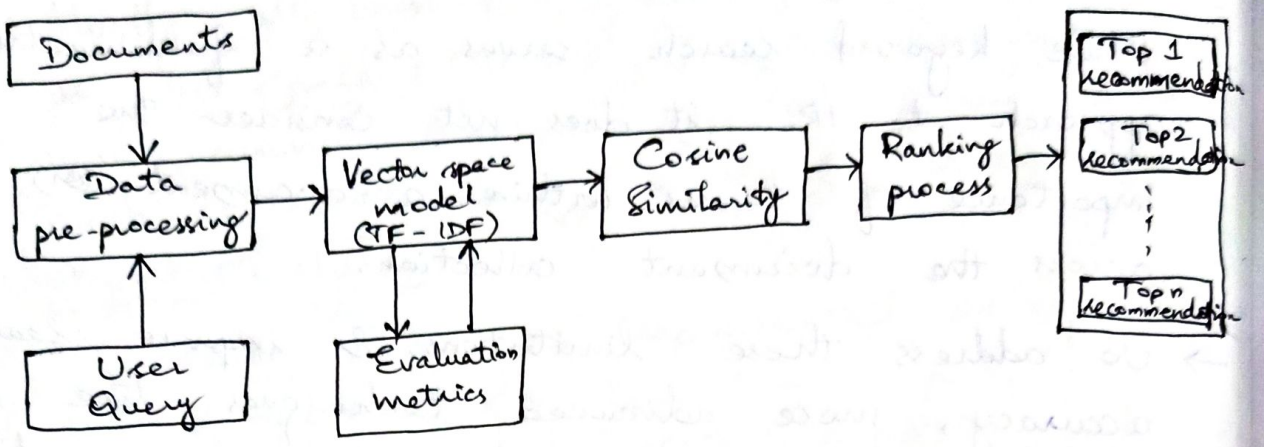
$$TF = \frac{\text{no. of times term 't' appears in a doc.}}{\text{total no. of terms in the doc.}}$$

Inverse Document Frequency (IDF) is the logarithmic ratio of total no. of docs. to that of no. of docs. with term 't' in it.

$$IDF = \log_2\left(\frac{\text{Total no. of doc.}}{\text{no. of doc. with term 't' in it}}\right)$$

→ TF measures the importance of a term within a specific document, whereas IDF is a measure of how common (or) rare a term is across the entire document collection.

Architecture of an IR process



Vector Space Model :-

→ is an algebraic model used in IR to represent documents & queries in a high-dimensional space.

→ Each dimension in this space corresponds to a unique term from the document collection, & the value of a vector in that dimension represents the importance of the term in corresponding document / query.

→ To represent docs. & queries as vectors, we first create a term document matrix (TDM), where each row represents a term & each column represents a document.

→ The value in the matrix at the intersection of a term & a doc. represents the importance of the term in that doc.

→ In case of TF-IDF model, this value is calculated using the TF-IDF formula.

→ Ultimately, a query is represented as a vector by transforming it into the same high-dimensional space as the documents, & the value is calculated using TF-IDF.

Cosine Similarity as a measure of document-query similarity:-

→ To measure the similarity b/w a document & a query in the vector space, we can use Cosine similarity & rank them accordingly.

Evaluation metrics:-

→ For each query, accuracy, precision, recall & F1 scores are calculated based on the no. of true positives, false positives, true negatives & false negatives.

		predicted +ve	predicted -ve
actual	+ve	true positive, t_p	false negative, f_n
	-ve	false positive, f_p	true negative, t_n

$$\text{Precision, } P = \frac{t_p}{t_p + f_p}$$

$$\text{Recall, } R = \frac{t_p}{t_p + f_n}$$

$$\text{Accuracy, } A = \frac{t_p + t_n}{t_p + t_n + f_p + f_n}$$

$$\text{F1 score, } F = 2\left(\frac{PR}{P+R}\right)$$

Ranking & Recommendation:-

→ Based on the cosine similarity scores obtained, ranking is done, which in-turn will be used to output the top recommendations with the order of highest priority on top of the list or first page of the search engines.

(5) TF-IDF Vectorization:-

→ Steps involved are:

(i) Clean data / Pre-processing
↳ normalize data (all lowercase), lemmatize data (all words to root words)

(ii) tokenize words with freq..

(iii) find TF for words

(iv) find IDF for words

(v) Vectorize vocabulary & interpret the results.

Eg:-

Doc 1 = Sun is shining hot today.

Doc 2 = Sun will not stop shining today.

Doc 3 = It will rain today night.

Step 1:-
Clean data + tokenize

Cleaned data	Count
be	3
hot	1
it	1
night	1
not	1
rain	1
shine	2
stop	1
sun	2
today	3

Step 2 → Find TF:-

Words	Doc 1	Doc 2	Doc 3
be	1/5 = 0.2	1/6 = 0.17	1/5 = 0.20
hot	1/5 = 0.2	0	0
it	0	0	1/5 = 0.20
night	0	0	1/5 = 0.20
not	0	1/6 = 0.17	0
rain	0	0	1/5 = 0.20
shine	1/5 = 0.2	1/6 = 0.17	0
stop	0	1/6 = 0.17	0
sun	1/5 = 0.2	1/6 = 0.17	0
today	1/5 = 0.2	1/6 = 0.17	1/5 = 0.20

$$TF = \frac{\text{no. of repetition of word in doc}}{\text{\# of words in a doc}}$$

Step 3 → Find IDF :-

Words	IDF value
be	$\log_2(3/3) = 0$
hot	$\log_2(3/1) = 1.59$
it	$\log_2(3/1) = 1.59$
night	$\log_2(3/1) = 1.59$
not	$\log_2(3/1) = 1.59$
rain	$\log_2(3/1) = 1.59$
shine	$\log_2(3/2) = 0.59$
stop	$\log_2(3/1) = 1.59$
sun	$\log_2(3/2) = 0.59$
today	$\log_2(3/3) = 0$

$$IDF = \log_2\left(\frac{\text{no. of docs.}}{\text{no. of docs. containing the word}}\right)$$

Step 4 → TF-IDF = TF * IDF :-

Words / Docs..	be	hot	it	night	not	rain	shine	stop	sun	today
Doc 1	0	0.318	0	0	0	0	0.118	0	0.118	0
Doc 2	0	0	0	0	0.27	0	0.1	0.27	0.1	0
Doc 3	0	0	0.318	0.318	0	0.318	0	0	0	0

Step 5 → Interpretation :-

→ for eg.., Doc 1 & Doc 2 informs something about 'shine', whereas Doc 3 has no relevance about it.

→ Doc 3 talks about 'night', & but no. info about the same in Doc 1 & 2.

... and so on.

Conclusion :-

→ The above table helps to find similarities & non-similarities b/w documents, words & more, much better than BoW.

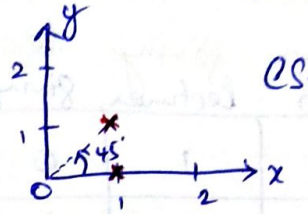
② Cosine Similarity :-

→ uses the cosine angle between the vectors.

→ measure of similarity between 2 non-zero vectors based on the inner product of cosine angles.

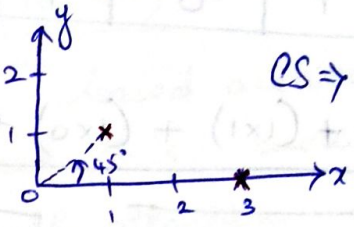
→ applications are to find text similarity in layer contents, eg :- Tweets of one topic.

	Hello	world
Hello World	1	1
Hello	1	0



CS ⇒ cos(45°) = 0.71

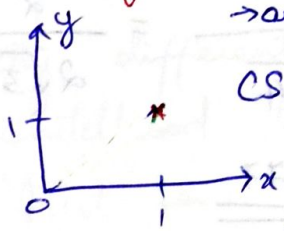
	Hello	World
Hello World	1	1
Hello Hello Hello	3	0



CS ⇒ cos(45°) = 0.71

→ Cosine Similarity ⇒ determined by the angle between 2 lines & not by lengths of the lines.

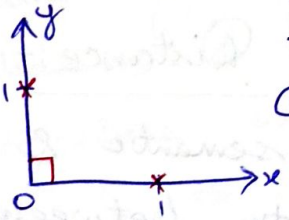
	Hello	World
Hello World	1	1
Hello World	1	1



→ angle is 0°

CS ⇒ cos 0° = 1

	Hello	World
Hello	1	0
World	0	1



→ angle is 90°

CS ⇒ cos 90° = 0

→ CS score ⇒ $0 \leq CS \leq 1$

- 0 → no overlapping (unique documents)
- 1 → exactly same docs.

Formula for CS with more than 2 documents :

$$= \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \cdot \sqrt{\sum_{i=1}^{n} B_i^2}}$$

A, B, ... ⇒ phrases / texts

Eg 1:-

$$CS \Rightarrow \frac{(1\times1)+(1\times0)}{\sqrt{1^2+1^2}\ \sqrt{1^2+0^2}} = \underline{0.71}$$

Hello World → Doc 1

Hello → Doc 2

A → Hello ; B → World

Eg 2:-

Doc 1 → I am a lecturer.

Doc 2 → I am giving lecture.

	I	am	lecturer	giving	lecture
A	1	1	1	0	0
B	1	1	0	1	1

$$CS \Rightarrow \frac{(1\times1)+(1\times1)+(1\times0)+(0\times1)+(0\times1)}{\sqrt{1^2+1^2+1^2+0+0}\ \sqrt{1^2+1^2+0+1^2+1^2}}$$

$$= \frac{1+1}{\sqrt{3}\sqrt{4}} = \frac{2}{2\sqrt{3}}$$

$$= \underline{0.577}$$

Similarity score will change when stop-word removal process is considered.

Machine Translation:-

Translation → activity [redacted], which involves translation of one human language to another.

Machine → done by machine.

Machine Translation → any computer software (AI source) translates one natural language into another target language automatically (without human intervention)

How does Machine Translation work?

⇒ At a fundamental level, it performs a straight-forward replacement of atomic words in a single characteristic language for words in another.

⇒ Using corpus methods, more complicated translations can be conducted, taking into account better treatment of contrasts in phonetic typology, express the acknowledgement & translation of idioms, just as the seclusion of oddities.

Benefits of Machine Translation:-

1) automated translation assistance
2) speed & volume ⇒ large swatches of text translated in a very short time.
3) large language selection
4) cost-effective translation

Application / Use cases of Machine Translation:-

1) internal communications
2) external communication
3) data analysis
4) online customer service
5) legal research

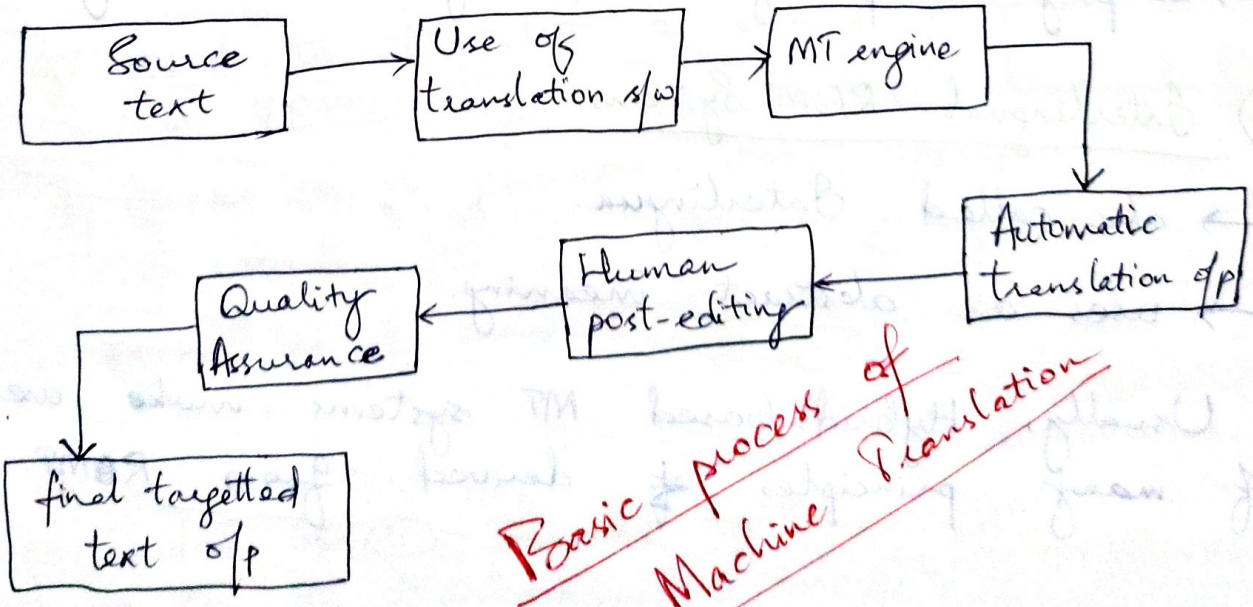
Types of Machine Translation:-

1) Rule-based MT
2) Statistical MT
3) Neural network based MT
4) Hybrid MT

→ All the above mtds. follow/work based on a 2-step process:

(i) decode the source language meaning of original text
(ii) encode the meaning into target language.

Original text/language → source language
language to be translated → target language



Basic process of Machine Translation

1) Rule-Based Machine Translation (RBMT):-

→ classical approach based on linguistic info. about source & target language, basically retrieved from unilingual, bilingual (or) multilingual dictionaries & grammars covering the main semantic, morphological & syntactic regularities of each language respectively.

→ Eg:- Systran, Japanese MT systems ⇒ old
Apertium, Grammatical Trans ⇒ new

Types of RBMT:-

1) Direct Systems (Dictionary-Based MT):-

→ commonly used for cross language retrieval systems.

→ generate the equivalent target query for the given source language query, i.e., map i/p to o/p with basic rules.

2) Transfer-Based MT:-

→ employ morphological & syntactical analysis.

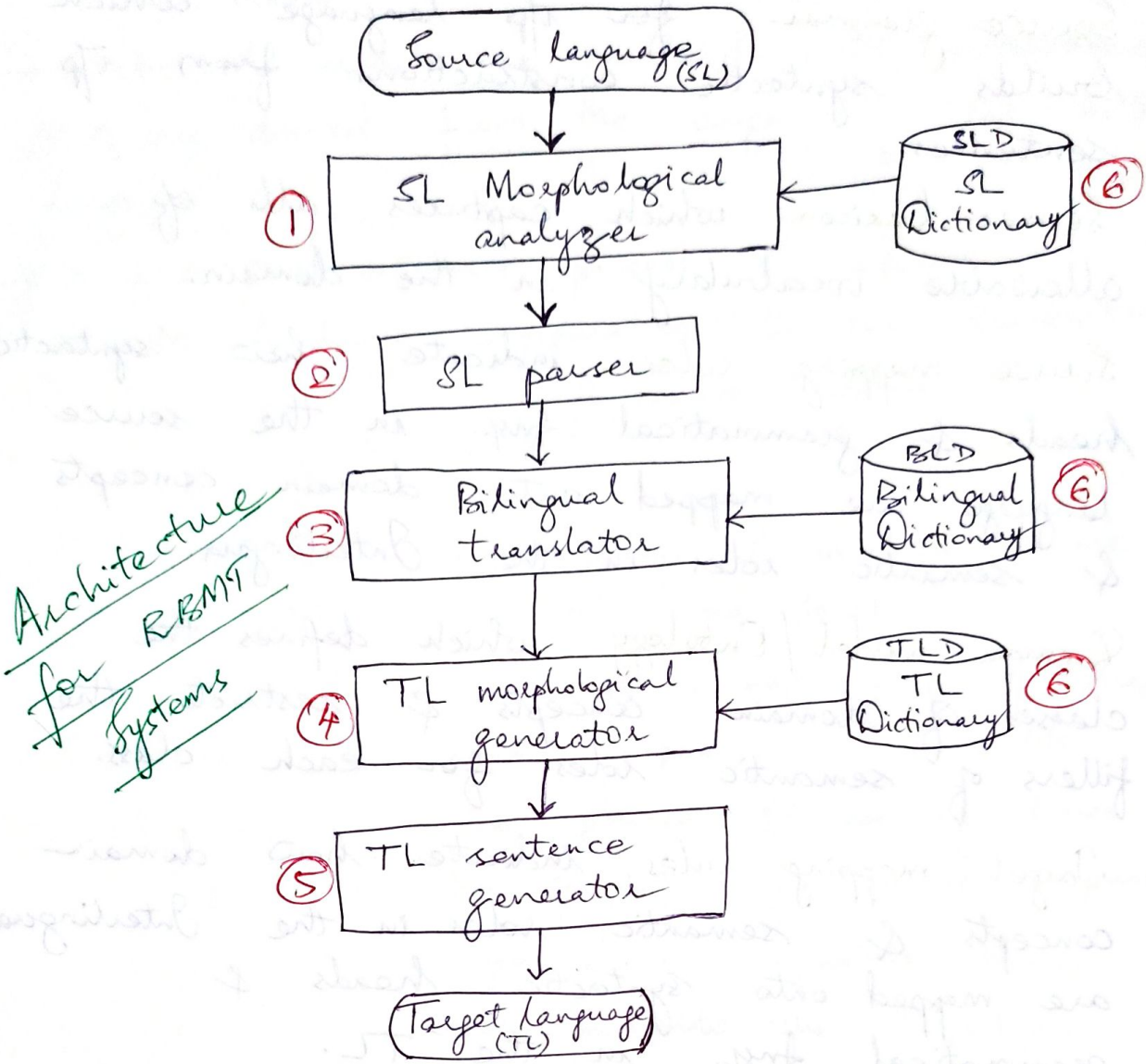
3) Interlingual RBMT Systems:-

→ also called Interlingua.

→ uses an abstract meaning.

Usually, Hybrid-based MT systems make use of many principles derived from RBMT.

Components of RBMT:-



Architecture for RBMT systems

① ⇒ analyses a SL word & provide the morphological info.

② ⇒ is a syntax analyser which analyses SL sentences.

③ ⇒ used to translate a source language word into TL.

④ ⇒ works as a generator for appropriate TL words, for the given grammatical info..

⑤ ⇒ works as a composer of suitable TL sentences.

⑥ ⇒ more specifically a mix. of 3 dictionaries, SLD → needed by ① for morphological analysis; BLD → used by ③ to translate SLW into TLW; TLD → needed by ④ to generate TLW.

RBMT makes use of the following:-

→ Source grammar for i/p language which builds syntactic constructions from i/p sentences.

→ Source Lexicon which captures all of allowable vocabulary in the domain.

→ Source mapping rules indicate how syntactic heads & grammatical fns.. in the source language are mapped onto domain concepts & semantic roles in the Interlingua.

→ Domain model / Ontology which defines the classes of domain concepts & restricts the fillers of semantic roles for each class.

→ Target mapping rules indicate how domain concepts & semantic roles in the Interlingua are mapped onto syntactic heads & grammatical fns.. in the TL.

→ Target Lexicon which contains appropriate target lexemes for each domain concept.

→ Target Grammar for TL which realizes target syntactic constructions as linearized o/p sentences.

Basic principles of RBMT & SMT with an example:-

* RBMT:-

→ main approach of RBMT systems is based on linking the structure of given i/p sentence with the structure of demanded o/p sentence, necessarily preserving their unique meaning.

Eg:- A girl eats an apple.

Source language → English

Demanded Target language → Deutsch (German)

→ at a minimum level, one needs the below entities to translate from a source language to the target language:

① a dictionary that will map each of the word in source language to an appropriate word in target language.

② rules representing regular sentence structure in both source & target languages.

→ In RBMT, the translation happens involving the following processes:

Stage 1:- tagging each word in source language with appropriate POS

a → article (indefinite) | eats → verb | apple → noun

girl → noun | an → article (indefinite)

Stage 2:- getting syntactic info. about the verb.

eat → present tense (simple)
→ 3rd person singular
→ active voice

Stage 3:- parsing the i/p sentence into NP & VP.

NP → A girl

VP → eats an apple.

Often only partial parsing is sufficient to get the syntactic structure of source sentence & to map it onto the structure of the target sentence.

Stage 4:- translation of words into target language (at word level only)

a → ein (category ⇒ indefinite article)

girl → Mädchen (category ⇒ noun)

eats → essen (category ⇒ verb)

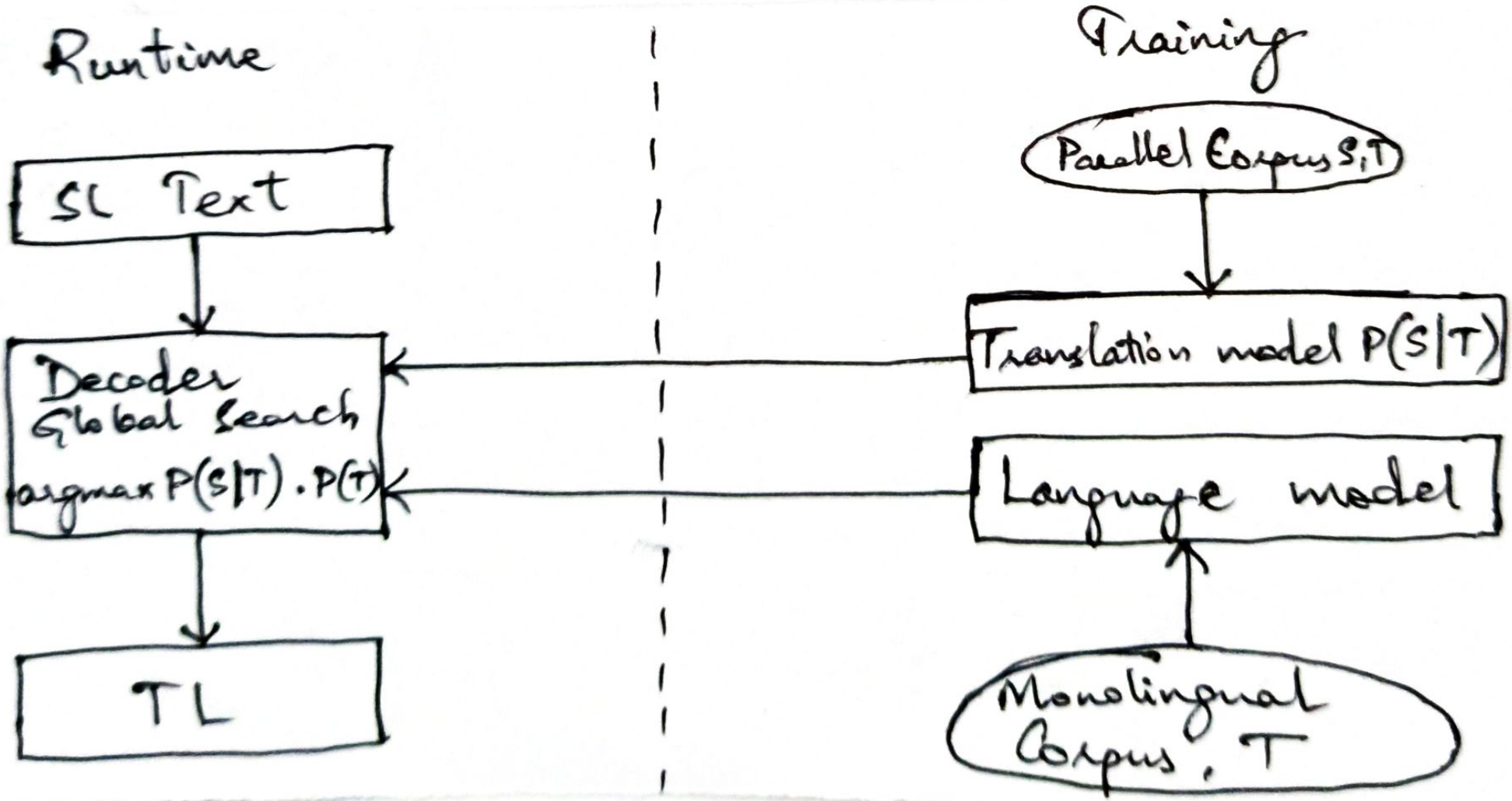
an → ein (category ⇒ indefinite article)

apple → Apfel (category ⇒ noun)

Stage 5:- mapping dictionary entities into appropriate inflected forms to reveal the final targetted o/p sentence.

A girl eats an apple ⇒ Ein Mädchen isst einen Apfel.

2) Statistical Machine Translation (SMT) :-



→ uses algorithms for learning how to analyze the human translations.

→ statistical models are involved with parameters that are derived from the corpora of the natural language.

→ it is a CPU intensive process & requires an extensive hardware configuration to run translation models for average performance levels.

Working principle :-

→ based on probabilistic theory & involves phrase level of translation & not word level.

→ translation model is built using freq. of phrases appearing the training corpus into a table.

→ This table stores the phrase & no. of times this repeats over the entirety of training corpus.

→ more frequently a phrase is repeated in training corpus, more probable the target translation is correct.

→ input ⇒ foreign sentence as an i/p.

→ output ⇒ generates a native sentence which is a translation of i/p.

→ language model ⇒ provides the probability of an arbitrary word sequence.

→ translation model ⇒ provides the probabilities of possible translation pairs.

→ decoding algorithm ⇒ graph search algorithm that provides best path on a word graph.

* SMT:-

→ main approach of SMT systems is based on translation of word phrases, unlike word-to-word translations in RBMT.

→ Grammar rules are not considered here & are often with low quality translation o/ps.

→ Pattern matching norms are considered here, however, the resultant o/ps are inconsistent between versions.

Eg:- A girl eats an apple in the morning.

Source language → English

Demanded target language → Deutsch (German)

→ In the source language English, the phrase "in the morning" can be placed in different positions of the sentence & the meaning still remains intact.

⇒ In the morning, a girl eats an apple.

⇒ A girl, in the morning, eats an apple.

⇒ A girl eats an apple in the morning.

→ However, the same concept cannot be followed in other languages, especially in German language.

→ As we have already seen, SMT does not know grammar norms, the o/p might be not accurate.

→ Considering the translation procedure, in phrases,

A girl	eats	an apple	in the morning.
ein Mädchen	~~essen~~	einen Apfel	am Morgen

→ Resultant o/p would be,

Am Morgen isst ein Mädchen einen Apfel. (or)

Ein Mädchen isst am Morgen einen Apfel.

→ The basic procedure would be as follows:

1) look for texts that are similar

2) break down into phrases / sentences

3) align the phrases/sentences / phrases within sentences.

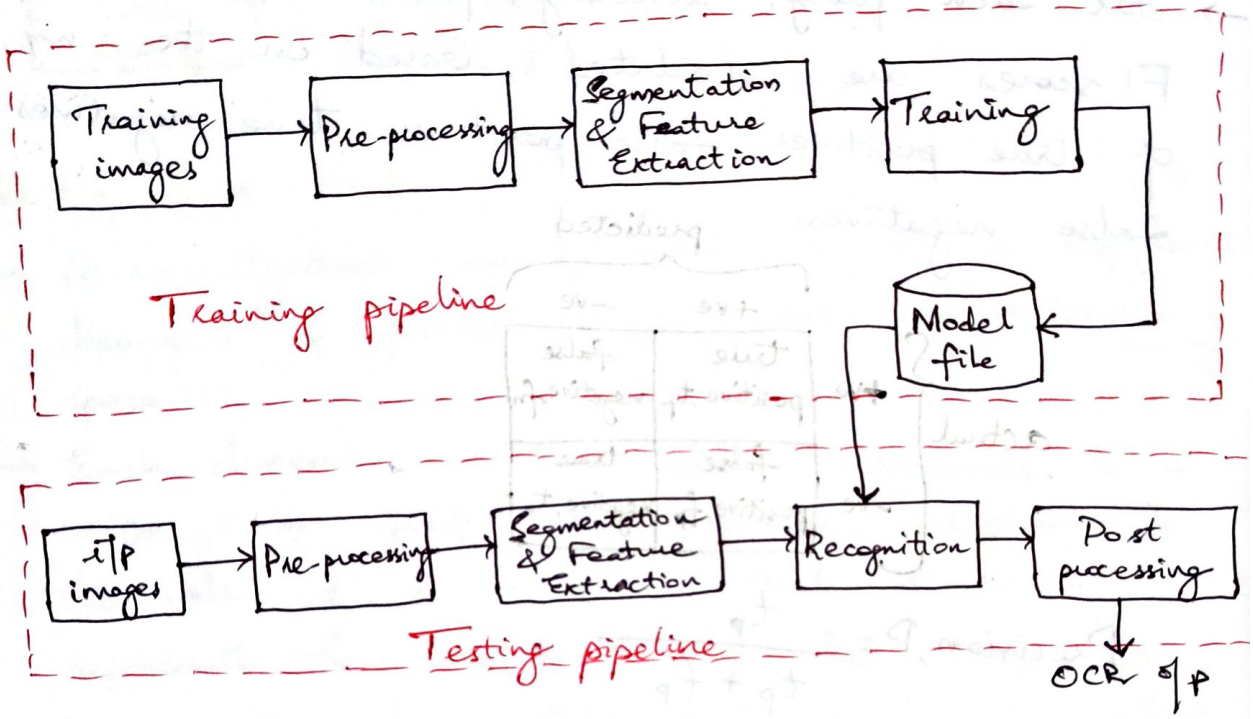
4) define distortion probability

5) Use Expectation - Maximization to improve the estimates, & iterate the procedure till convergence.

Optical Character Recognition (OCR):-

→ it is the process of detecting & reading text in images through computer vision, which enables NLP algorithms to decipher the text & make sense of what the document conveys.

How does a generic OCR work?:-



→ OCR algorithms can be based on traditional image processing & ML/DL-based methods.

→ Traditional approaches to OCR go through a series of pre-processing steps where the inspected document is cleaned & made noise-free.

→ Then the doc. is binarized for subsequent contour detection to aid in the detection of lines & columns.

→ Finally, characters building the lines are extracted segmented & identified via various ML algorithms like KNN & SVM.

→ Algorithms that combine Vision & NLP - based approaches have been particularly successful in providing superior results for text recognition & detection.

→ Generally, OCR methods include vision - based approaches used to extract textual regions & predict bounding box co-ordinates for the same.

→ The bounding box data & image features are then passed onto Language Processing algorithms that use RNNs, LSTMs & Transformers to decode the feature - based info.. into textual data.

→ DL - based OCR algorithms have 2 stages:

(i) Region proposal :-

→ involves the detection of textual regions from the image, which is achieved by the convolutional models that detect segments of text & enclose them in bounding boxes.

→ These regions are used as attention maps & fed to language processing algorithms along with features extracted from the image.

(ii) Language Processing :-

→ NLP - based networks like RNNs & Transformers work to extract info.. captured in these regions & construct meaningful sentences based on features fed from the CNN layers.

Text Summarization:-

→ one of the most effective & simplest technique for giving the central idea (or) main information from the large amount of information.

→ NLP expresses the interaction b/w human language & computer, & Automatic Text Summarization (ATS) is a subset of NLP that uses advanced algorithms & ML models to analyze & condense lengthy texts into smaller, digestible paragraphs (or) sentences.

→ applications are in academia, business, news, legal areas, healthcare, etc.

Common ATS Techniques:-

(1) Input-based ATS:-

→ **Single-Document Summarization** involves summarizing content from a single document.

→ **Multi-Document Summarization** involves summarizing content from multiple documents, which is more complex due to the need to understand relationships b/w different texts.

(2) Output-based ATS:-

→ **Extractive Summarization** selects & isolates essential information from the original text to create a summary.

→ **Abstractive Summarization** generates a new, unique summary that captures the main ideas of the original text using NLG.

(3) Purpose-based ATS:-

→ **Generic Summarization** provides an overview of the main points in a text without specific assumptions about the content.

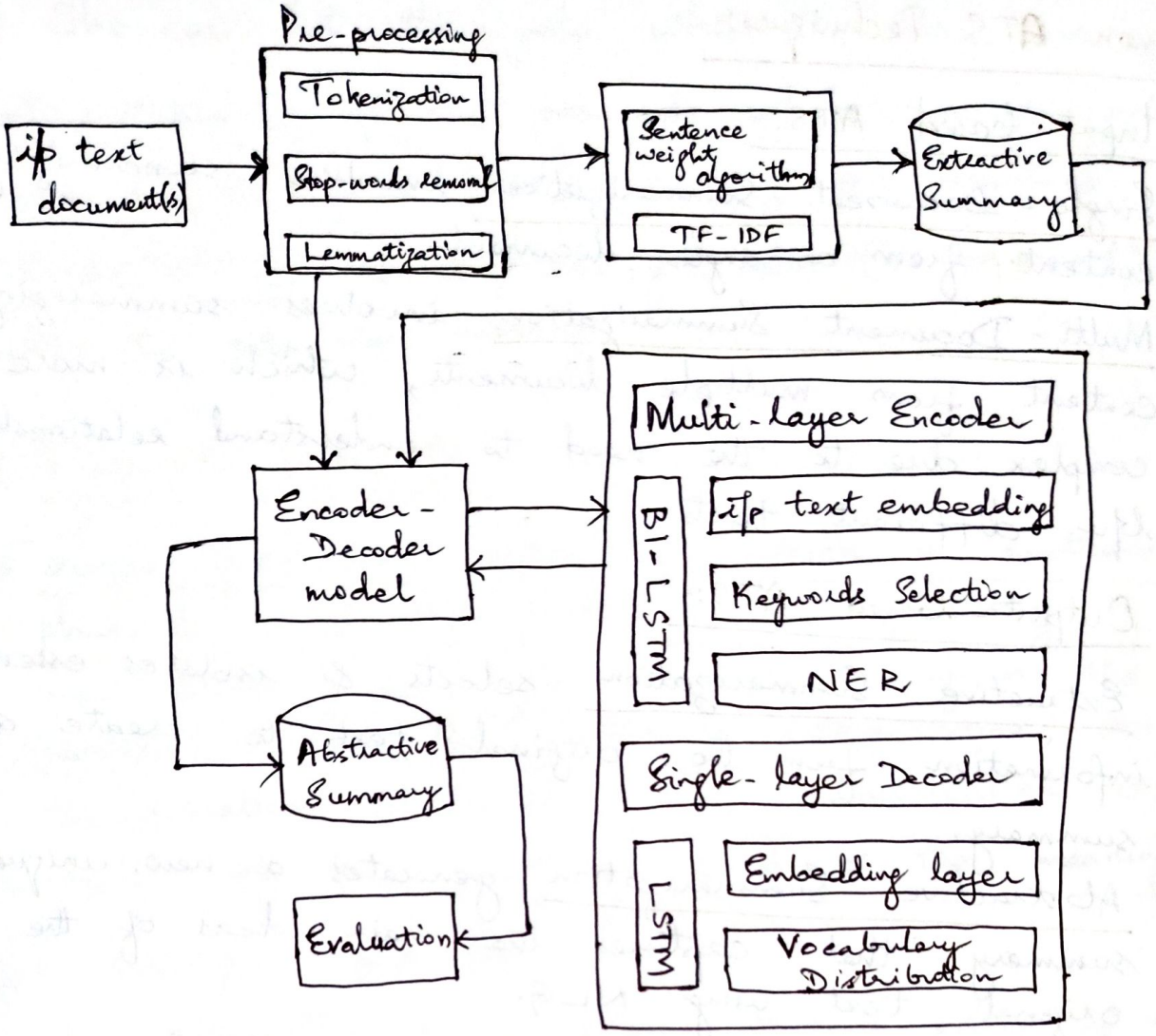
→ **Domain-specific Summarization** uses domain-specific knowledge to generate tailored summaries.

→ **Query-based Summarization** generates summaries that answer specific questions about the text.

Architecture of Extractive & Abstractive ATS:-

→ The i/p documents are initially completed with pre-processing which involves tokenization, stop-word removal & stemming.

→ After pre-processing, next step is to map each sentence into a fixed length vector of real nos.s which is known as sentence encoder.



→ After pre-processing, the characteristics are retrieved & sentences are sorted according to their load & the freq. with which each phrase (or) token occurs.

→ For feature extraction, extractive summarization selects pertinent phrases based on several feature sets to provide an extractive summary for capturing the core concept of the innovative text.

→ The weights are assigned to the source phrases & the heavily weighted sentences are assigned a higher rank for a summary generation.

→ Sentence weight algorithms are used for important sentence retrieval, a statistical method depending on the weights given to each phrase.

→ The statistical technique recognized as TF-IDF helps in identifying which terms are common than others.

→ The sentences are then ordered from lower to higher TF-IDF values & focus is placed on selecting the sentences with a high TF-IDF value for a comprehensive summary.

→ For generating abstractive summary, the encoder-decoder paradigm is employed when one sequence serves as the contribution & another as the o/p.

→ The derived / generated summary is produced by using Bi-LSTM on the encoder side & LSTM on the decoder side.

→ The produced abstractive summary is then evaluated using usual (or) other special forms of evaluation metrics related to accuracy, precision, recall, F1-score, etc.